Performance testing

Date	30 September 2026
Team id	SWTID-2026-2551
Project name	Fit Buddy – AI Fitness Plan Generator using Gemini Models
Maximum mark	4 marks

Performance Testing

Performance testing evaluates how the **Fit Buddy** system behaves under expected and peak user load. The testing focuses on the application response time, throughput, error rate, CPU utilization, and memory utilization while generating personalized fitness plans using the Gemini AI model.

Step 1: Testing Overview

Field	**Details**
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	AI Fitness Plan Generation API
Test Environment	Local Development Environment
Test Date	30 September 2026

The template asks for the testing tool, testing type, target module/API, environment, and test date.

Step 2:test scenarios

S.No	**Test Scenario / Description**	**No. of Virtual Users**	**Duration (sec)**	**Expected Outcome**
1	User Login API Performance Test	10	60	Login requests should be processed successfully with minimal delay.
2	Fitness Profile Submission Test	10	60	Fitness profile data should be submitted and

				processed successfully.
3	AI Fitness Plan Generation Test	10	120	Gemini should generate personalized fitness plans within an acceptable response time.
4	Multiple Concurrent Plan Generation Test	20	120	System should handle multiple plan-generation requests without critical errors.

This follows the four scenario rows and fields provided in the original template.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	**[Enter measured value]**	**[Pass/Fail]**	Average API response time
2	Response Time (Max)	< 5 seconds	**[Enter measured value]**	**[Pass/Fail]**	Maximum response time recorded
3	Throughput (Req/sec)	**[Enter target]**	**[Enter measured value]**	**[Pass/Fail]**	Requests successfully processed per second
4	Error Rate	< 1%	**[Enter measured value]**	**[Pass/Fail]**	Percentage of failed requests
5	CPU Utilization	< 80%	**[Enter measured value]**	**[Pass/Fail]**	CPU usage during performance testing
6	Memory Utilization	< 80%	**[Enter measured value]**	**[Pass/Fail]**	Memory usage during testing

The response-time, error-rate, CPU, and memory targets above follow the values explicitly provided in your template.

Important: The template does not provide actual measured performance values, so those values should be filled in after you run the tests. This avoids claiming test results that have not actually been measured.

Step 4: Observations & Analysis

The Fit Buddy application was evaluated under multiple simulated user requests to assess the performance of its authentication, fitness profile, and AI fitness-plan-generation functionality. Particular attention was given to the AI Fitness Plan Generation API because it communicates with the Gemini model.

The test results should be analyzed based on:

- Average and maximum API response time.
- Number of requests successfully processed.
- Percentage of failed requests.
- CPU utilization during concurrent requests.
- Memory utilization during testing.
- System behavior when multiple users request fitness plans simultaneously.
- Gemini API response time and availability.

If all measured values remain within the defined target limits, the corresponding test cases can be marked **Pass**.

Step 5: Screenshots / Evidence

Attach screenshots of the performance-testing results below, such as:

1. Postman API test results.
2. AI Fitness Plan Generation API response.
3. Response-time measurements.
4. Number of successful and failed requests.
5. CPU and memory utilization.
6. Any performance-testing report generated during testing.

The original template specifically requests screenshots/evidence from the performance-testing tool, such as JMeter, Locust, or k6 output.